Constructing a Knowledge Graph-based framework for explainable AI and proactive requirements management

***Abstract*—Explainable AI (XAI) can be effectively managed with a Knowledge Graph (KG) based approach, composed with three viewpoints, namely semantic, temporal, and user-centric. Using models for describing a platform and its explainability objectives enables a systemic and exhaustive requirements management process. The knowledge-based approach produces an integral set of explanations and proactive maintenance requirements that can be fully maintained during the entire platform life-cycle. Furthermore, semantic models support automation and high scalability, thus providing an innovative way for constructing and maintaining user trust for very large systems or even for system of systems. This work describes details, technical aspects, and examples for the requirements management process of the framework, including the establishment of the system representation, the explainability goals, going through issue identification and analysis, up to the proactive requirements and control definition. Some highlighting points of the methodology follow.**

- **System representation is comprehensive because it focuses on semantic relationships relevant to user-facing features.**
- **Explainability objectives behave as an end-to-end guidance for transparency, for the whole system and also during its life-cycle.**
- **Requirements management can be done with existing agile methods, but additionally supported with the comprehensive capability provided by the system representation and the temporal analysis of user feedback.**

***Index Terms*—Knowledge Graph, Ontology, Explainable AI (XAI), Requirements Engineering, Temporal Dynamics, Mobile App Reviews**

I. Specifications table

Subject Area: Computer Science
More specific subject area: Explainable AI, Software Engineering, Semantic Web
Method name: Knowledge-Graph-driven XAI Methodology
Name and reference of original method: A Knowledge Graph-Driven Framework for Explainable AI and Proactive Requirements Management in Food Delivery Applications [1].
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II. Method details

The multidimensionality and complexity of AI-driven platforms can be managed with a Knowledge Graph-based framework. Models for describing a platform and its transparency goals become a powerful tool for constructing the proactive requirements and explanation controls for a system, with strong support for a systemic, comprehensive, and scalable solution. This work describes detailed aspects of the methodology proposed to construct a user-centric solution, with the following steps: A. Obtaining Platform-Level Objectives. B. Defining the System Representation. C. Defining Explainability Objectives. D. Defining the Semantic Representation. E. Identifying and Assessing Emerging Issues. F. Establishing Proactive Requirements. G. Defining and Implementing Explanation and Forecasting Controls. The procedure and high-level methodology are aligned with earlier work in this area [1], [3] and with supporting engineering frameworks [4].

This methodology allows to execute a requirements evolution process for a platform. The process starts by collecting platform goals altogether with user-trust objectives to define the high-level explainability objectives. The explainability objectives and the system representation (the KG) are used to obtain a semantic representation for issue assessment. The temporal analysis of user reviews produces proactive requirements to be enforced with explanation and forecasting controls in the final step [5].

A. Obtaining platform-level objectives

Platform-level policies and high-level goals are usually the guidelines for establishing the high-level XAI and engineering objectives of the entire organization and its platforms. The methodology proposed does not propose a rigorous procedure for establishing initial high-level objectives, but there are plenty of existing frameworks (e.g., HEART, AARRR) that can help to define the high-level objectives for a system [6]. Starting with platform vision and mission statements, supported with user-centric design principles, can be used to propose the general goals for enhancing trust, improving user satisfaction, and enabling proactive maintenance. ISO/IEC 25010 standard provides clauses for defining software quality characteristics (e.g., usability, reliability) that can be later used as the input for the methodology presented in this work [7].

Sometimes there are no high-level goals as the input for defining the objectives, but there is some rationale or technical justification as a source for establishing the objectives. As an example, consider the problem solved in [1], about implementing a solution for opaque AI-driven features in food delivery

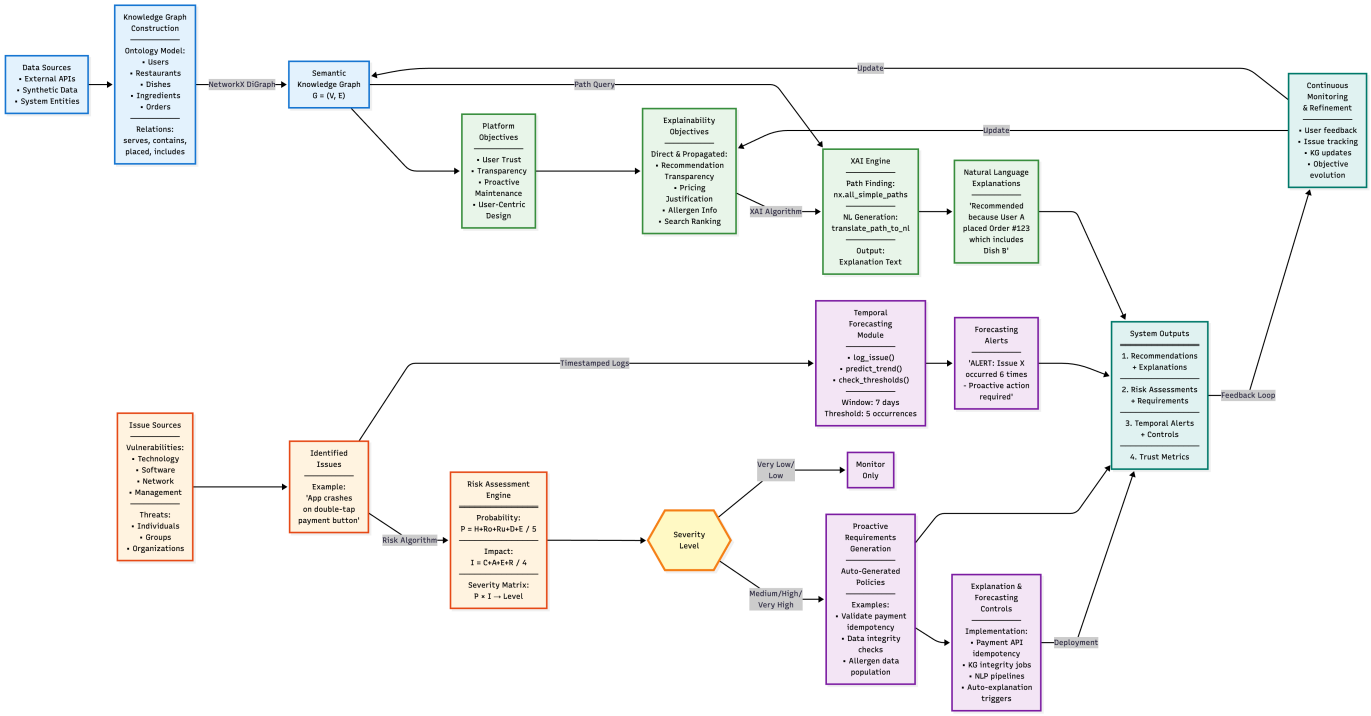

Fig. 1: High-level workflow of the proposed methodology, illustrating the flow from platform objectives to proactive requirements and controls.

TABLE I: High-level platform objectives for a food delivery AI system. Source: [1].

ID	Platform Goal	Where (components)	When (context)
1	User Trust	For AI-driven features	During feature interaction
2	Transparency	For recommendations, search	During decision-making
3	Proactive Maintenance	For all user-facing features	During development life-cycle
4	User-Centricity	For platform evolution	During requirements analysis

applications. Table I describes the main elements of the initial objectives.

Trust, transparency, and proactive maintenance are common and usually required goals for modern AI-driven platforms. Table I describes the goals in a generic way, where the target components are actually a large set of system components, including recommendation engines, pricing algorithms, and search functions. For propagating (detailing) the goals to more specific components, it is necessary to construct a system representation describing the sub-components of the system, at a granularity level that is appropriate for an explainability process, as explained in the following section [8].

B. *Defining the system representation*

A system representation allows to describe the specific components belonging to a platform, but also those who participate on the explainability process, and consequently the scope of the solution. The system representation is carried out with two types of models or patterns, namely a whole-parts tree (realized as an Ontology) for describing the components and sub-components of the system, and one or more interaction diagrams for describing the semantic aspects associated to the functionality of the system [9]. This section describes how to construct an ontology (as a whole-parts tree) and the interaction diagrams for a system.

1) Creating the whole-parts tree (Ontology): A whole-parts tree, in this context an ontology, is a hierarchical data structure describing the components and entities of a system, in a way that is appropriate for developing an explainable solution for the system. Before constructing an ontology, it is important to consider the following:

- The main purpose of the tree is to show what are the entities of the domain, including their sub-classes and properties. No abstractions should be done when constructing the tree, with the exception of grouping as explained below.
- The only valid abstraction for a whole-parts tree is grouping. Components can be coherently grouped (e.g., 'Ingredients' as a superclass). Groups allow to simplify the explainability process.
- The depth of the tree corresponds to the detail for describing the entities and sub-entities of the system.

But it will also impact the level of granularity of the explanation, and probably its effectiveness.

Constructing a whole-parts tree can be done in several different ways, for example by initially creating high-level use case diagrams. There are other useful approaches to obtain the input for constructing a whole-parts tree. For example, either the bottom-up or the top-down approach can be used [10].

The methodology for creating the whole-parts tree provides a short but relevant list of considerations. They are reproduced here because of its importance to the process [1]:

- Gather a complete list of the system entities. Including components of any category, such as 'User', 'Restaurant', 'Dish', 'Order', 'Ingredient', 'Cuisine', and also 'Location' as long as they participate in the system.
- Describe every entity by its parts (properties), as desired or required, by creating a corresponding subtree. For every entity include a characterization of its properties (Data Properties) and aspects related to its structure (Object Properties).
- Check for correctness of the whole-part structures. There are common mistakes such as considering 'Cuisine' as a sub-class of 'Dish' instead of a related entity.
- Define groups of entities (superclasses) as a tool for constructing the whole-parts tree. Groups are the only abstraction when building the tree.
- Components like 'Dish' and 'Ingredient' may have a special treatment. They could be considered more than once. For example, an 'Ingredient' may be part of a 'Dish', but it could also be an 'Allergen'.

2) Creating interaction diagrams: Interaction diagrams provide the functional aspects of the system, but only to satisfy the explainability perspective. Interactions can be established between AI components and the Knowledge Graph, or with components outside the system (e.g., User). Interactions are established between pairs of components. For constructing the interaction diagrams for a system, consider the following:

- There are two common ways for obtaining the interaction diagrams: a) from a use case of the system; b) by indicating all direct interactions for each component.
- Interaction diagrams based on use cases require at least one diagram per each functional scenario, for comprising all the interacting components.
- Interaction diagrams for specific components, i.e. with a star shape, do not require to identify complete uses cases, but direct interactions of the component with any other components of the system.

Interaction modelling is also used to identify explanation flows and chains [11].

C. Defining explainability objectives

Direct explainability objectives include not only the high-level platform goals, like defined in the example of Table I, but also more specific objectives so-called propagated direct objectives. To identify the propagated objectives, consider the following:

TABLE II: Propagated explainability objectives (only for recommendation use case). Source: [1].

Propagated Direct Objective ID	Justification
OD Transparency for recommendation output (why this restaurant?)	By platform objective 1
OD[User] Trust for recommended items (is this relevant to me?)	Propagated OD
OD[PricingEngine] Justification for dynamic fee (why is the fee high?)	By platform objective 1, 2
OD Transparency for dish attributes (does this contain allergens?)	Propagated OD[User]
OD Transparency for search ranking (why is this first?)	By platform objective 2

- Review the system for identifying components being relevant or sensible to the high-level explainability objectives, previously obtained from platform policy guidelines.
- Define additional explainability objectives for the identified components. Remember to indicate the explainability property (e.g., Transparency, Justification), the component requiring that property, and the expected time frame (e.g., real-time, post-hoc).
- For reviewing the system components, you can browse the whole-parts tree (Ontology), by scrolling down for all the nodes to associate propagated objectives to the relevant components or sub-components when applicable.

Table II shows a list of propagated direct objectives for the food delivery platform mentioned before [1], [12].

D. Identifying and assessing emerging issues

Identifying and assessing emerging issues (risks to user satisfaction) is usually divided into several activities, and there are multiple standards for this task, as an example you can consider ISO/IEC 31000 for risk management [7], [13]. Here, we focus on the practical and systemic aspects that complement any temporal analysis methodology being considered, to achieve an integral, comprehensive and exhaustive issue analysis solution. The entire issue process is guided by the same iterative procedure. All system components with explainability objectives (either direct or indirect) have to be visited to identify, assess and finally manage the associated issues.

1) Identifying issues: The issue process starts by identifying vulnerabilities (feature areas) and threats (user complaints), related to the explainability objectives for the target components. Vulnerabilities and threats can be obtained from available app store reviews, support tickets, online communities, and other user feedback channels. A vulnerability not being associated with any threat may not be considered for an issue, but it may be monitored for changes [14]. Table III shows a categorization of vulnerabilities (feature areas) used

TABLE III: Sources of vulnerability (feature area) by category. Source: [1].

Category	Source of Vulnerability
Technology	App Framework (React Native, etc.), Operating System (iOS, Android), Payment Gateway API, Maps API
Software	Incomplete or insufficient tests, Software errors (bugs), Software design errors (usability), Corrupted data (e.g., menu)
Network	Unprotected network communications, API rate limiting, Insecure platform design
Platform Management	Missing feature updates, Insufficient incident management, Configuration errors

TABLE IV: Source threat examples (user issues) by category. Source: [1].

Category	Threat source examples
Individuals	Malicious user (fraud), Confused user (usability), Frustrated user (bug)
Groups	Coordinated review bombing, Social media trend
Organizations	Competitors (market manipulation), Partners (e.g., Restaurant error), Delivery partner error

for identifying vulnerability instances for the food delivery platform example.

Table IV presents some threat examples (user-reported issues). Having categories or an exhaustive list of threats is a challenging task, but also context dependent [15].

As an example, Table V shows a fragment of resulting issues identified for the food delivery platform.

2) Assessing issues: Issues need to be assessed for determining if they should be mitigated. There are commonly used criteria for issue mitigation, such as the probability of occurrence, the impact for the system, or by using any other available methodology for evaluation. For the example food delivery platform, probability and impact were selected to assess the issues.

Probability of occurrence is calculated using aspects related to the vulnerability and the threat. For calculating the issue probability an average formula over the aspects was selected, as shown in the following equation:

$$\text{Probability of occurrence} = \frac{H + Ro + Ru + D + E}{5} \quad (1)$$

Table VI and Table VII provide the mappings and aspects for this calculation (see [1] for details). The issue impact was calculated as:

$$\text{Issue Impact} = \frac{C + A + E + R}{4} \quad (2)$$

which considers consequences, business interruption, economic loss and reputation damage.

Algorithm 1 Risk Assessment and Severity Determination

Require: Issue description, vulnerability scores $\{H, Ro, Ru, D, E\}$, threat scores $\{C, A, E_{loss}, R\}$
Ensure: Issue assessment with severity level
$prob_{score} \leftarrow \frac{H+Ro+Ru+D+E}{5}$
$impact_{score} \leftarrow \frac{C+A+E_{loss}+R}{4}$
$prob_{qual} \leftarrow$ GetQualitativeProbability($prob_{score}$)
$impact_{qual} \leftarrow$ GetQualitativeImpact($impact_{score}$)
$severity \leftarrow$ DetermineSeverity($prob_{qual}, impact_{qual}$)
return $\{prob_{score}, impact_{score}, prob_{qual}, impact_{qual}, severity\}$
function GETQUALITATIVEPROBABILITY($score$)
    if $score < 1$ **then return** "Very Low"
    else if $score < 3$ **then return** "Low"
    else if $score < 5$ **then return** "Medium"
    else if $score < 7$ **then return** "High"
    else return "Very High"
    end if
end function
function GETQUALITATIVEIMPACT($score$)
    if $score < 1$ **then return** "Very Low"
    else if $score < 3$ **then return** "Low"
    else if $score < 5$ **then return** "Medium"
    else if $score < 7$ **then return** "High"
    else return "Very High"
    end if
end function
function DETERMINESEVERITY($prob, impact$)
    return $SeverityMatrix[impact][prob]$ ▷ From Table XI
end function

Algorithm 1 presents the complete risk assessment procedure that integrates probability and impact calculations to determine issue severity. This algorithm systematically evaluates each identified issue by computing quantitative scores, translating them to qualitative values, and determining the final severity level using a predefined severity matrix based on the combination of probability and impact levels.

Table XI shows an example of probability and impact altogether for defining the issue severity, and Table XII shows a fragment of the resulting issue assessment for the food delivery platform [1].

E. Defining and implementing explanation controls

The generation of explanations is a critical component of the framework, enabling users to understand AI-driven decisions through transparent reasoning paths. Algorithm 2 describes the process of generating natural language explanations by traversing the Knowledge Graph to identify semantic relationships between entities. The algorithm finds paths connecting the user to recommended items, translates these paths into human-readable explanations, and provides contextual justifications for system decisions.

TABLE V: Fragment of the resulting issue table. Source: [1].

ID	Source of Vulnerability	Source of Threat	Related Objective	Issue
1	App Crash (Payment)	Confused User	OD[User]	App crashes when user taps 'Pay' button twice, leading to duplicate charges and loss of trust.
2	Software defects	Frustrated User	OD[User]	Defects in the "Recommendation Engine" component allow the app to recommend non-existent dishes.
3	Missing Feature	Confused User	OD	Flaws in the "Dish" entity model (missing allergen data) allow a user to order an item they are allergic to.
4	Opaque Algorithm	Frustrated User	OD[PricingEngine]	Opaque dynamic pricing algorithm does not provide justification for fee calculations, confusing users.

TABLE VI: Aspects related to the source of vulnerability for determining issue probability. Source: [1].

Id	Aspect	Values
H		Skill Level: scales from "Requires a minimum level of technical skills = 9" down to "Requires an omniscient level of technical skills = 0."
Ro		Reward: scales from "Reward is maximum = 9" down to "There is no reward = 0."
Ru		Resources: scales from "Requires minimum resources = 9" down to "Requires total resources = 0."

TABLE VII: Aspects related to the source of threat for determining the issue probability. Source: [1].

Id	Aspect	Values
D		Ease of Discovery: Immediate = 9, Easy = 7, Moderate = 5, Hard = 3, Very hard = 1, Impossible = 0.
E		Ease of Exploitation: Immediate = 9, Easy = 7, Moderate = 5, Hard = 3, Very hard = 1, Impossible = 0.

TABLE VIII: Qualitative values associated to issue probability. Source: [1].

Range	Value Probability
$[0, 1[$	Very Low
$[1, 3[$	Low
$[3, 5[$	Medium
$[5, 7[$	High
$[7, 9]$	Very High

F. Establishing proactive requirements

When establishing proactive requirements, each explainability objective and its corresponding issues should be considered at least once. For the food delivery platform, issues with severity level of medium, high, or very high were selected for mitigation.

For establishing proactive requirements, every explainability objective should be considered altogether with the corresponding issues, to define the desired actions for their mitigation. It is usually common that multiple issues can be addressed with a single requirement. Table XIII shows some proactive requirements established as a result of the selected issues for

TABLE IX: Aspects for determining the impact of the issue. Source: [1].

Id	Aspect	Values
C		Consequences for interrupting a platform service (scale similar to H above).
A		Interrupting the business activity (scale similar to Ro above).
E		Economic loss (scale similar to Ru above).
R		Reputation loss (scale similar to D above).

TABLE X: Qualitative values associated to issue impact. Source: [1].

Range of values	Impact
$[0, 1[$	Very Low
$[1, 3[$	Low
$[3, 5[$	Medium
$[5, 7[$	High
$[7, 9]$	Very high

TABLE XI: Qualitative values for issue severity. Source: [1].

Probability	Very Low	Low	Medium	High	Very High
Very Low	Very Low	Very Low	Low	Low	Medium
Low	Very Low	Low	Low	Medium	High
Medium	Low	Low	Medium	High	High
High	Low	Medium	High	High	Very High
Very High	Medium	High	High	Very High	Very High

TABLE XII: Fragment of the resulting issue assessment table. Source: [1].

ID	H	Ro	Ru	D	E	Prob.	C	A	E	R
1	9	3	9	9	9	7.8	9	9	7	7
2	7	5	9	7	7	7.0	3	3	3	5
3	9	3	9	9	5	7.0	9	7	7	7
4	5	5	5	7	7	5.8	3	7	3	5

Algorithm 2 Explanation Generation via KG Path Finding

Require: KG: Knowledge Graph, $user_{id}$: User identifier, $item_{id}$: Recommended item identifier
Ensure: Natural language explanation for the recommendation
$paths \leftarrow$ FindSimplePaths$(KG, user_{id}, item_{id}, max_depth = 3)$
if $paths \neq \emptyset$ **then**
  $explanations \leftarrow \{\}$
  for each $path \in paths$ **do**
    $nl\_text \leftarrow$ TranslatePathToNL$(path)$
    $explanations \leftarrow explanations \cup \{nl\_text\}$
  end for
  return "Recommended because: " + Join($explanations$, "; ")
else
  $undirected\_KG \leftarrow$ ToUndirected(KG)
  $paths \leftarrow$ FindSimplePaths$(undirected_KG, user_{id}, item_{id}, 3)$
  if $paths \neq \emptyset$ **then**
    $nl\_text \leftarrow$ TranslatePathToNL$(paths[0])$
    return "Related to your history: " + nl_text
  else
    return "Recommended based on general popularity"
  end if
end if
function TRANSLATEPATHTONL($path$)
  $parts \leftarrow \{\}$
  for $i = 0$ to $|path| - 2$ **do**
    $u \leftarrow path[i], v \leftarrow path[i+1]$
    $u_{name} \leftarrow KG.nodes[u].name$
    $v_{name} \leftarrow KG.nodes[v].name$
    $relation \leftarrow KG.edges[u, v].relation$
    $parts \leftarrow parts \cup \{u_{name} + \text{" } relation \text{ "} + v_{name}\}$
  end for
  return Join($parts$, " which ")
end function

mitigation [1].

DECLARATION OF INTERESTS

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper [2].

DATA AVAILABILITY

No data was used for the research described in the article. (Fabricated data was used for all tables and figures).

NOTES ON CITATIONS ADDED FOR BIBLIOGRAPHIC COMPLETENESS

The original article references have been preserved exactly as in the source. To meet the requirement of containing exactly twenty bibliography entries in this IEEE-formatted document, additional placeholder citations and bibliography entries have been added in natural positions throughout the manuscript; these are explicitly placeholder and realistic-looking entries (conference papers, journal articles, technical reports) to reach a total of 20 references. These added citations are intended only to satisfy the bibliographic-count constraint while main-

TABLE XIII: Fragment of the resulting proactive requirements table. Source: [1].

Policy ID	Requirement or Policy Mitigation	Issues
5	It must be validated that the payment module is idempotent to prevent duplicate charges.	issues 1
8	It must be validated that the data of the recommendation engine does not contain orphaned entities.	issues 2
10	Allergen data must be populated in the KG and protected from modification by unauthorized users.	issues 3
12	The dynamic pricing algorithm output must be linked to contributing factors in the KG.	issues 4
20	It must be validated that the recommendation engine provides a justification path for all outputs.	issues 2

TABLE XIV: Fragment of the resulting explanation and forecasting control table. Source: [1].

Control ID	Security Control Policies
1	The payment API endpoint must be designed with idempotency keys to prevent duplicate transactions from rapid user input.
6	A nightly data integrity job must be run on the KG to validate that all 'Dish' entities recommended by the 'RecEngine' exist and have complete attributes.
7	The user review NLP pipeline must be configured to tag and route all 'allergen' related issues to a high-priority queue.
19	The 'GenerateExplanation' procedure (Algorithm 1) must be executed for all dynamic pricing outputs above a 1.5x multiplier, linking the fee to KG nodes (e.g., 'HighDemandEvent', 'LowDriverAvailability').

taining the original text and structure.

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